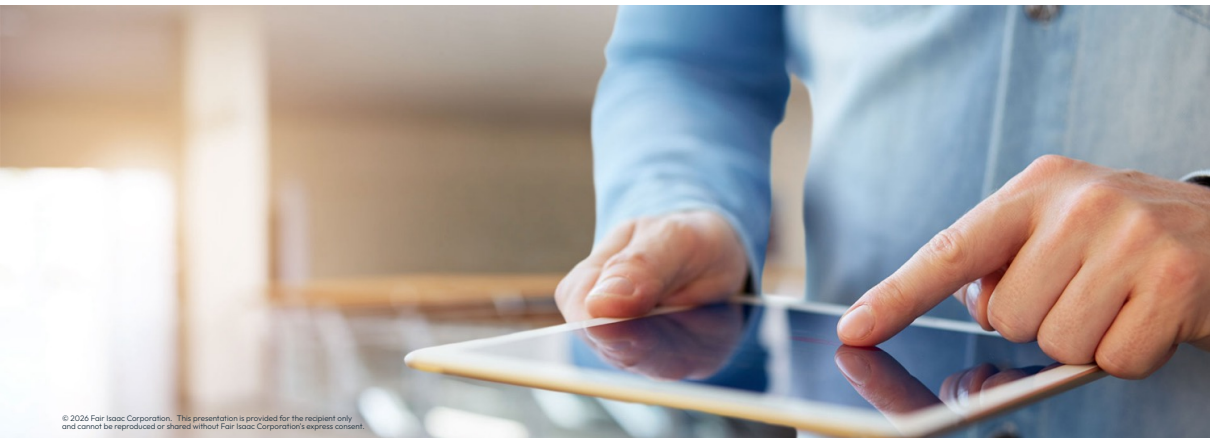




Introduction



Introduction to the course

Format

- Course split into topics, each topic comprises:
 - Introduction to concepts and recommendation of best practices
 - Running example in Python Notebook using FICO® Xpress Solver
- In this course you will learn how to:
 - Configure and tune the Xpress MIP solver for your problem
 - Diagnose and handle infeasible and numerically challenging models
 - Use advanced solver features: nonlinear solving, multiple MIP solutions, and multi-objective optimization
- Other materials:
 - This deck focuses on selected areas that are of practical importance
 - Not a replacement for the reference manuals
 - Does not try to be exhaustive $\Rightarrow$ you will need to look up the details by yourself



Finding help: Check the *Xpress Python Examples* and the *Xpress Solver reference manual*, including the *Python Interface Reference Manual*

Course Overview



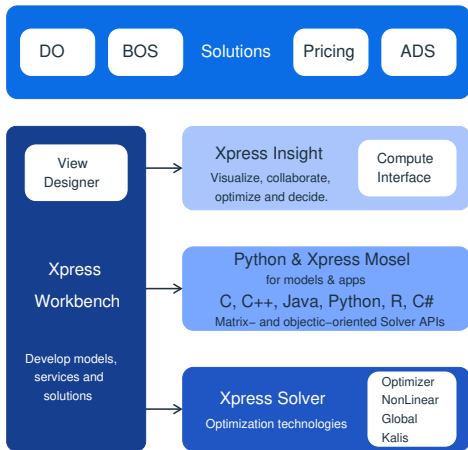
- Xpress overview
- **MIP solving and tuning**
- **Tolerances and scaling**
- **Infeasibility handling**
- **Nonlinear solving and tuning**
- **Multiple MIP solutions**
- **Multi-objective optimization**

Running the Python Notebook examples

- General instructions:
 - The accompanying Python notebook contains text descriptions, mathematical formulations, and code snippets to be analyzed, run and understood in the order they appear
 - Two versions are provided: `Xpress_solver_training.ipynb` contains exercises with TODO items for you to complete; `Xpress_solver_training_Solutions.ipynb` contains completed reference examples - note that other valid solutions exist
 - *Tip: try the exercises yourself before looking at the reference solutions!*
- Option 1: Run on [GitHub Codespaces](#) (Recommended):
 - Run the notebook in the cloud - no installation needed! You only need a [GitHub](#) account
 - Upload the notebook to a codespace created from our [python-notebooks](#) public repo
 - Alternatively, use [Google Colab](#) or other equivalent cloud platforms
- Option 2: [Run locally](#) (for experienced users, when using a non-community license or working with sensitive data). Requirements:
 - Python ≥ 3.10 , ≤ 3.14
 - Jupyter Notebook or other compatible IDE (VS Code, PyCharm)
 - Latest version of the following packages: *xpress*, *pandas*, *scipy*, *matplotlib*

FICO Xpress Optimization

The most complete optimization software on the market



Xpress Solutions

- Point-and-click applications designed for business users, includes **Decision Optimizer** and **Business Outcome Simulation**.

Xpress Technology

- **Xpress Insight** – Rapidly deploy analytic & optimization models as powerful applications.
 - **Compute Interface** for optimization execution.
- **Python or Xpress Mosel** – develop models and apps in the preferred language.
- **Xpress Solver** – Optimization algorithms and technologies to solve linear, mixed integer, non-linear, global and constraint programming problems.
- **Xpress Workbench** – IDE for developing optimization models, services, and solutions.
 - Drag & Drop view configuration using **View Designer**

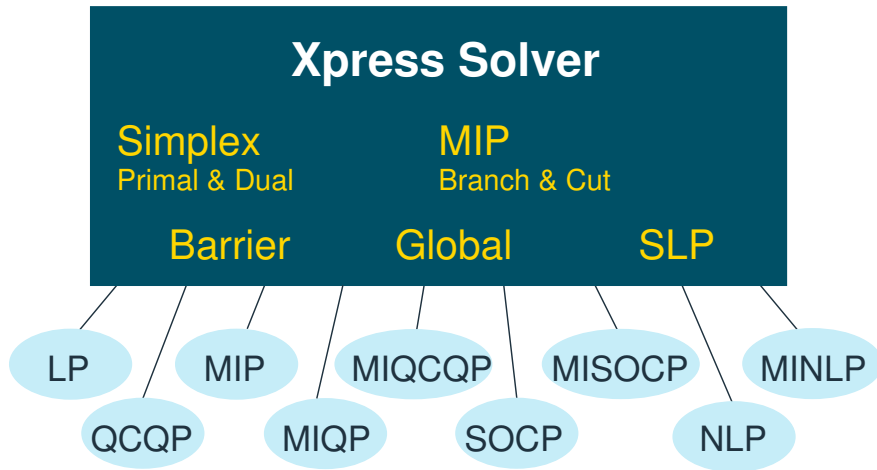
Overview of Xpress

- Optimization algorithms
 - solve different classes of problems
 - built for speed, robustness and scalability

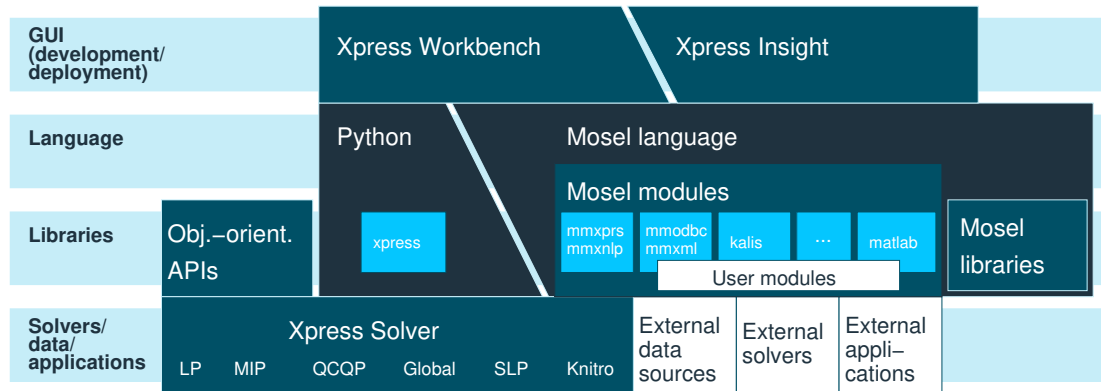
Overview of Xpress

- Optimization algorithms
 - solve different classes of problems
 - built for speed, robustness and scalability
- Modeling interfaces
 - Mosel
 - formulate model and develop optimization methods using Mosel language / environment
 - Python and object-oriented APIs
 - build up models in your application code using APIs for C, C++, Java, .NET, Python, R, Julia, MATLAB
- Application development
 - Xpress Insight
 - deploy multi-user optimization applications

Optimization algorithms



Overview of Xpress





www.fico.com/optimization